Coursework for Algorithms for Data Analysis and Data Mining (I+II)

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A. Data Analysis and Regression

1 Logistic Regression

Data provided for analysis with logistic regression was placed in a table below.

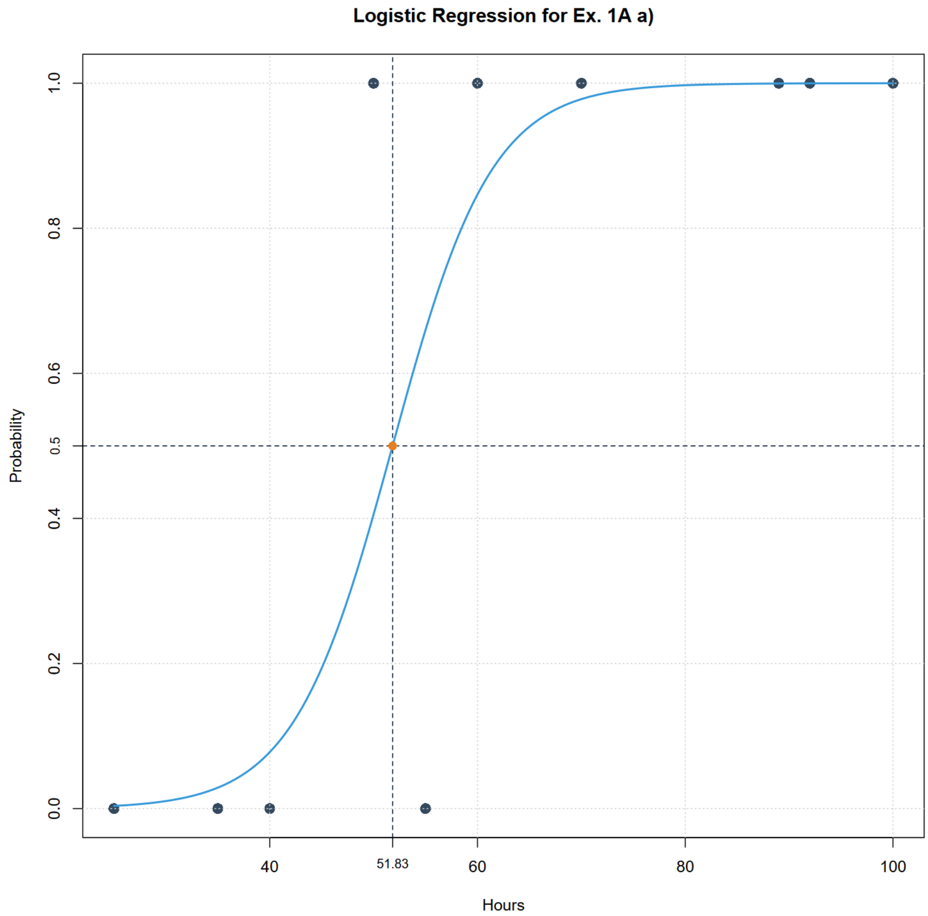
Table 1: Numbers of hours on programming and final result.

Hours	25	35	40	50	55	60	70	89	92	100
Result	0	0	0	1	0	1	1	1	1	1

a) For this exercise we used code from file "Zad1Aa.R".
Summarizing the logistic regression for provided data set we receive following parameters:

Variable	Estimate	Std. Error	z value	p-value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

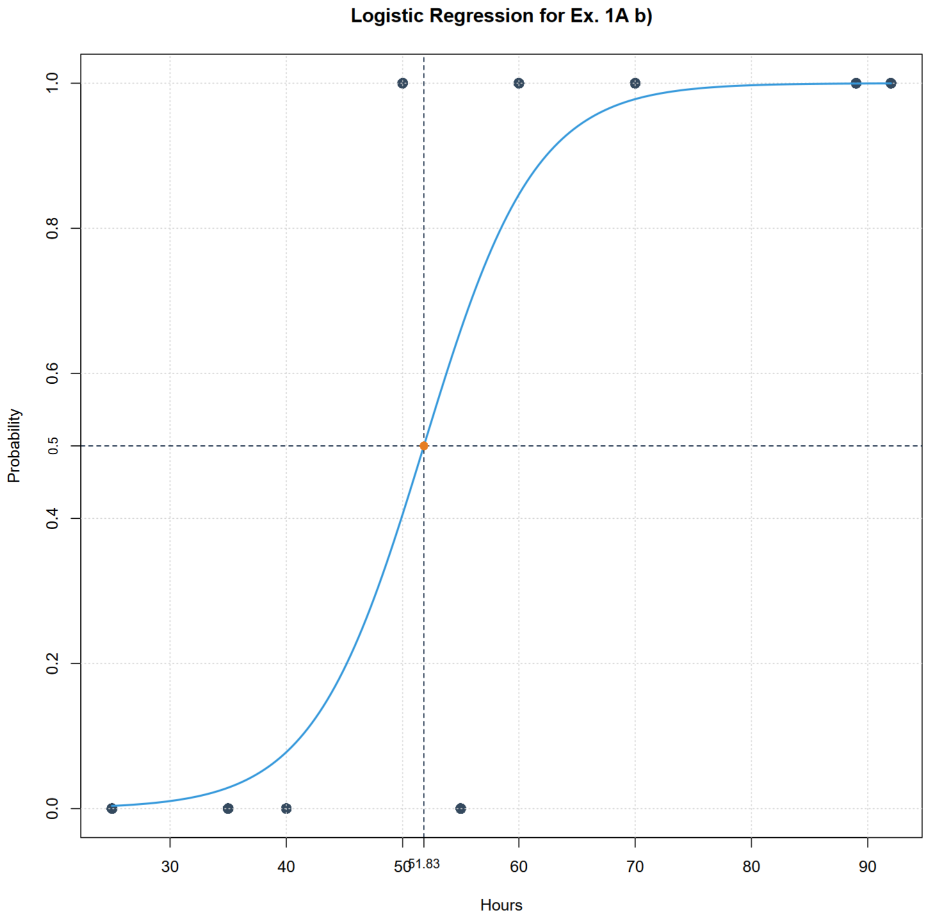
As we may see, basing on data from table 2, the z values for both variables are rather high what is proven by p-value being higher than 0.1 what can be considered as quite large. It is indeed result of low amount of data - 10 records equivalent to 9 degrees of freedom. This means that significance of our model is low thus statistically it is not important. Nevertheless, we may conclude that direction of dependence between hours of programming and final result is positive.
We may also plot the graph below.



b) For this exercise we used code from file "Zad1Ab.R".
Summarizing the logistic regression for provided data set, excluding point (100, 1), we receive following parameters:

Variable	Estimate	Std. Error	z value	p-value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

As we may see value of all parameters is the same as from the a) sub-point.

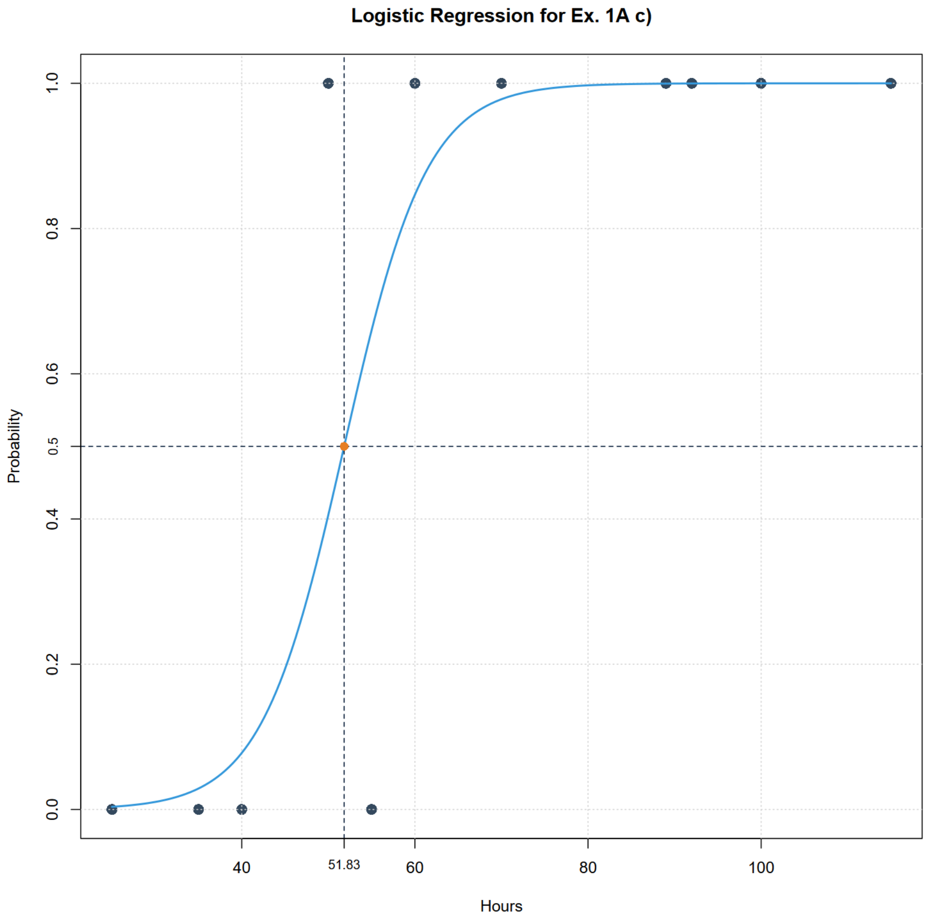


Moreover, the graph is also the same. Thus, we may conclude that removing point (100, 1) does not influence this model. We may suppose that it influenced by almost perfect correspondence of coordinates of this point with proposed model.

c) For this exercise we used code from file "Zad1Ac.R".
Summarizing the logistic regression for data set from sub-point a, including new point (115, 1), we receive following parameters:

Variable	Estimate	Std. Error	z value	*p*-value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

As we may see value of all parameters is the same as from the a) and b) sub-points.



Moreover, the graph is also the same. This is not surprising as it confirms hypothesis from sub-point b) that when coordinates is almost perfectly corresponding with our model. This we may conclude, that the model would only change if value on y axis were 0 which we would then consider as outlier. But ones placed on x axis to the right and keeping y=1 does not affect model.

d) To sum up all above we may propose advantages and disadvantages

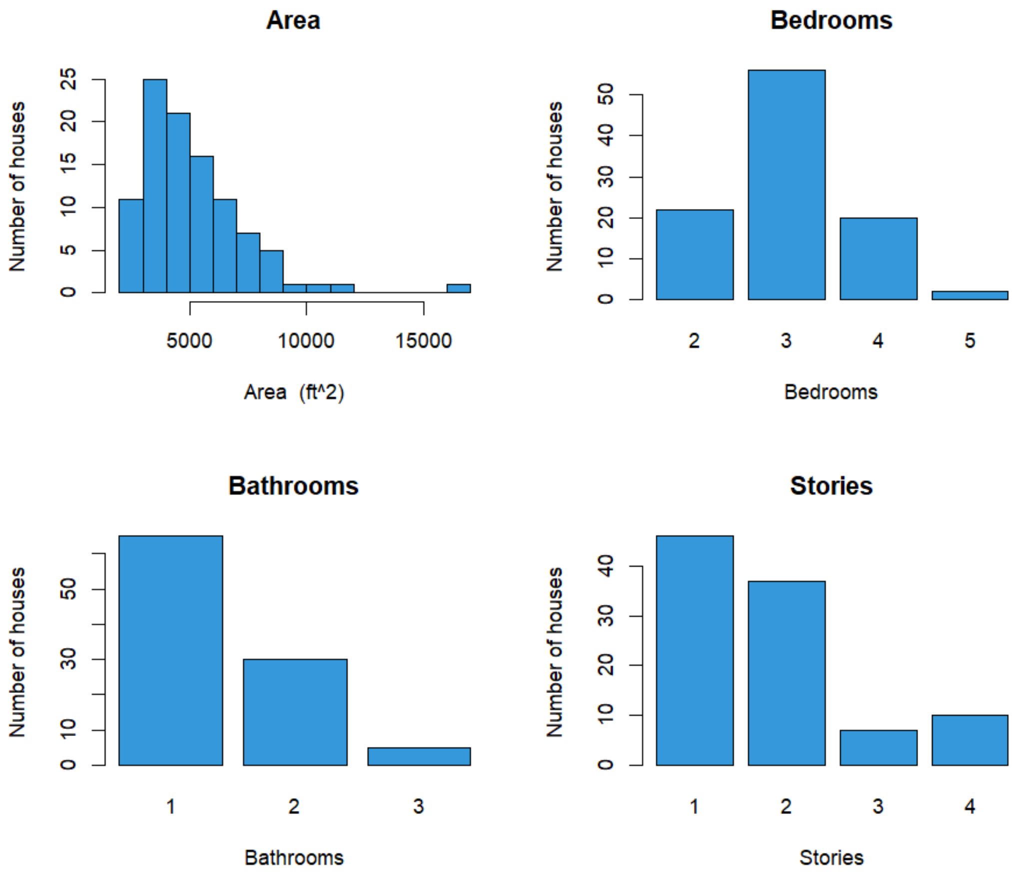
Advantages	**Disadvantages**
Simple algorithm	Very sensitive to outliers

B. kNN Regression Analysis

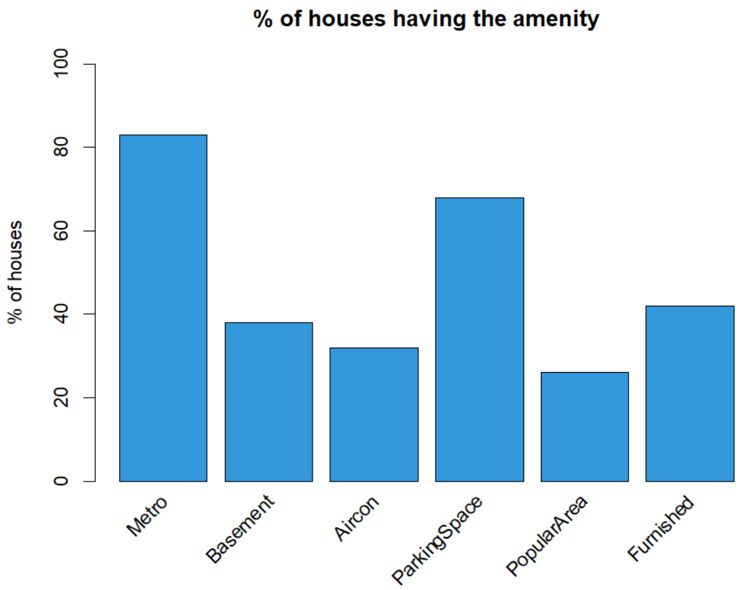
1 Summary of the sggw.b data

Here we used code from Zad2.R file. We will provide statistical analysis for the data. We have 10 explanatory features and one target feature - the price of the house. In the explanatory features (X-features) we have the Area, the numbers of Bathrooms, Bedrooms etc, and binary feautures telling us if there is metro in close proximity, is the place furnished, or is it a popular area.

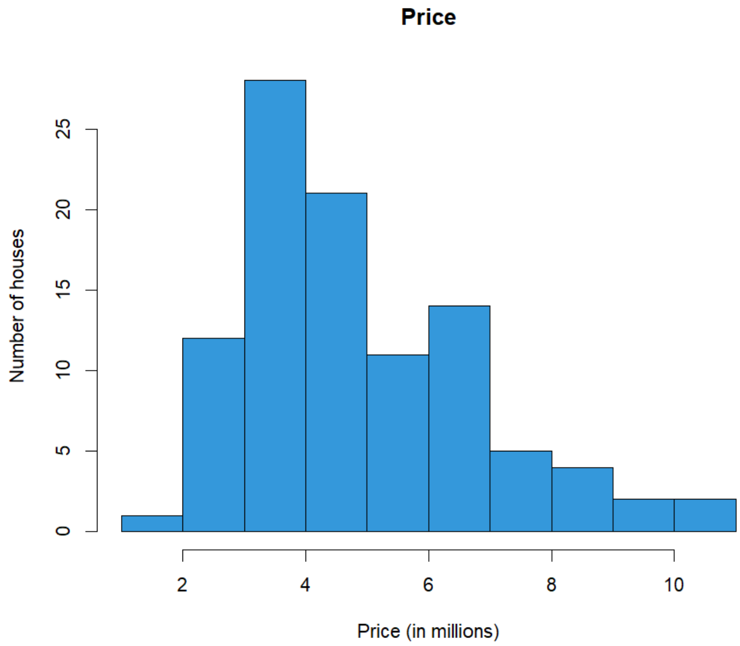
	Mean	SD	Min	Max
Area in ft^2	5167.32	2186.64	2145.00	16 200.00
Bedrooms	3.02	0.71	2.00	5.00
Bathrooms	1.40	0.59	1.00	3.00
Stories	1.81	0.95	1.00	4.00
Metro	0.83	0.38	0.00	1.00
Basement	0.38	0.49	0.00	1.00
Aircon	0.32	0.47	0.00	1.00
ParkingSpace	0.68	0.87	0.00	3.00
PopularArea	0.26	0.44	0.00	1.00
Furnished	0.42	0.50	0.00	1.00
Price (in mln.)	4.88	1.93	1.75	10.85



We can see that the area is similar to the χ^2 distribution, the peak is around 3000^2 feet. Most houses have 3 bedrooms, 1 bathroom and 1 or 2 stories



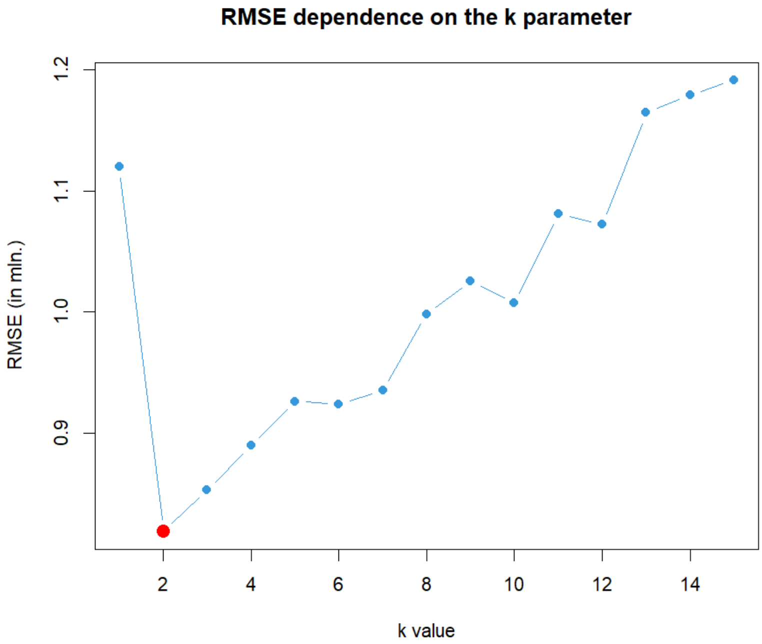
Most of the houses has a metro station nearby at over 80 %, and a parking space at 70 %. The place being furnished or having a basement has around 40 % chance. The least chance is to get an air conditioning or being in the popular area.



The price has a distribution similar to the area distribution.

2 kNN regression data analysis

Why did we choose kNN? It is perfect for predicting the value of the houses, as it chooses the most similar places, and checks the price with them. We used FNN (Fast Nearest Neighbor Search Algorithms and Applications) library, in which it is possible to do regression on k nearest neighbours model. We need to scale the data, since the kNN alghoritm uses Euclidean metric of proximinity. We firstly split the data into train and test set (80/20), and then trained the scaler on training data. The next step was to scale the test data, but the scaling was based only on the center and scale of the training data, to avoid bias. Then we tried to find the number of neighbours which minimizes the RMSE value. It turned out, that the best value was *k = 2*.



The RMSE is 25% of the mean price. It is an acceptable value, since we had only 100 houses to train a model on. It is interesting how only the 2 neighbouring houses had given the best results in our analysis. It can mean that the variance between the data points was high. The R^2 is 82,6%, which is a very good result. Mean average error is 0.623 mln, and the mean absolute percentage error is 11.1%. So the average error of the prediction was only 11% of the actual price. Given the simplicity of the alghoritm, this result is a good result.